11 October 2026

The Editors
PLOS ONE

Re: Research Article submission, "Allowance edges or transfer edges? Evaluating on-chain smart contract reputation out of window against degree baselines"
Running title: *Allowance versus transfer edges for smart contract reputation*

Dear Editors,

We submit the above manuscript for consideration as a Research Article in *PLOS ONE*. Before a user trades through a router or deposits into a vault, the user signs an ERC-20 approval that lets that smart contract move the user's tokens. The manuscript asks what these approvals, recorded on chain as allowances, add to graph reputation scores for smart contracts compared with the token transfers on which existing scores are built. On public data from the Arbitrum One rollup we compare PageRank on the latest allowances (EndorseRank) with the published transfer-based Adaptive Weighted PageRank (AWP) and with the raw degree counts that both walks smooth, for 15,107 contracts, and we freeze every score before the outcomes it is tested on.

Why the work matters beyond its case study. Wallets, protocols and security tools must decide which contracts deserve spending authority, and graph reputation scores are one proposed aid. Our results show why such scores should be tested out of window and against the counts they smooth. The transfer walk beat its count in two windows, whereas the allowance walk with uniform restarts fell below the plain count of approving owners, in comparisons made after the labels were known; a closed form shows that for most contracts it is a reweighted approval count. On an outcome that neither graph contains, the later liquidation avoidance of contracts trading on a perpetual futures exchange, the comparison registered before this study extracted the July–September outcomes did not separate the two edge types, in the registered analysis or on the revised scores. The evaluation design and the code are public.

Why *PLOS ONE*. The journal assesses scientific soundness rather than perceived impact and publishes null and negative results. Several of our central findings are negative or qualified: EndorseRank with the planned uniform restarts did not beat the approval count, the comparison on trader outcomes was not resolved, and a variant that did beat the count was found after the labels were known and is reported as a hypothesis. We also report a deviation openly: after the results of the registered analysis were known we revised how token amounts are valued, from raw units to US dollars over five widely traded tokens. The manuscript reports the revised analysis, applies the registered rules to it as a re-analysis, sets the registered results beside it, and gives the plan, the registration and the revision verbatim, with their commits and hashes, as supporting information. Every effect size comes with its paired bootstrap interval.

Statements. The manuscript is not under consideration elsewhere and has not been published. It is derived from the first author's doctoral research at the University of South Africa (UNISA). A related manuscript by the same authors, on a walk that mixes the two edge types, the decision rules fixed for it and the Sybil exposure of the allowance walk, has been submitted to *IEEE Access*; we upload it as a related manuscript file. The two papers share the data, the cohorts, the registration and the revision, and both report the spender holdout coefficients of the raw degrees, EndorseRank and AWP with their degree contrasts, the closed form of EndorseRank and the solver runtimes. The allowance-against-transfer comparison on trader outcomes is reported only in this manuscript; on the same trader cohort the related manuscript reports only the contrasts of its mixed walk that its own registered rule names. The code, configuration, queries, result summaries, per-contract scores and labels, and pre-specification documents are publicly available at `https://github.com/abitocodes/phd_works` and `https://github.com/abitocodes/phd-dissertation-draft-en`. The study uses only public blockchain data and public token prices and involves no human participants; only contracts are scored. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors received no specific funding for this work. Claude (Anthropic) was used to write and run analysis code, generate tables and figures, and draft and revise the text; the full statement is in Methods (Use of generative AI).

Thank you for considering our work.

Yours sincerely,

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